HEINZ NIXDORF INSTITUT
UNIVERSITÄT PADERBORN

Foundations of Knowledge Graphs
Introduction to exercises

Summer Term 2026

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Overview

Contents today

Overview

- Organization
  - Repetition of the lecture

- Exercises
- Mini-Project

PANDA

- TL;DR: Find everything in **PANDA**
- Find it here: dice-research.org/teaching/ → https://panda.uni-paderborn.de/course/view.php?id=69971

Section 1

Organization

Organization

Time, Location, Format

- Format = 3 + 2
  - 3 SWS of lecture
  - 2 SWS of exercise

Organization

Time, Location, Format

- Format = 3 + 2
  - 3 SWS of lecture
  - 2 SWS of exercise
- Lectures: Wednesday 09:00-11:00 Room F0.530
- Exercises: Friday 12:00-14:00, (bi-weekly) Room F2.211
- Exercises are not graded

Organization

Exam

- Format
  - Exam: Written
  - Content: Lecture + Exercises
  - Duration: 120 minutes
  - Target Date: TBA (tentative)
  - 2nd Exam: TBA
  - Location: TBA

Section 2

Exercises

Organization

Exercises (subject to change)

- Bi-Weekly from today onwards
- First appointment is to clarify the submission system
- 14 days for completion of each series
- Submitted by Thursday latest at 09:00am (time stamp of our server)

Organization

Exercises

- Requirements
  - Course content
  - Working knowledge of Java or Python
- FAQ and discussion forum
  - Intended for feedback and discussions
  - Not for sharing solutions to exercises
  - PANDA
- Web exercises in Jupyter
  - Work with your favourite IDE
  - Integrate your results in Jupyter for submission

Exercises

Proposed workflow

1. Exercise is online
2. Read the exercise (**carefully!**)
3. Solve the exercise using your IDE
4. Test your exercise with the provided test cases
5. Copy your solution into the Jupyter UI
   Make sure it is working in Jupyter!
6. Submit your solution in Jupyter/Panda

Exercises

Important hints

- Read the exercise description carefully!
  Test your implementation regarding corner cases.

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- Make sure that your solution has a low runtime complexity. Your file has only 30 seconds (sometimes up to 5 minutes).

Exercises

Important hints

- Read the exercise description carefully!
  Test your implementation regarding corner cases.
- Don't wait until the last moment for submission!
- Make sure that your solution has a low runtime complexity. Your file has only 30 seconds (sometimes up to 5 minutes).
- If you encounter issues with Jupyter, contact us (e.g., create a post in the forum in PANDA or (in special cases) write an mail to `saleem@mail.uni-paderborn.de` with `[Knowledge Graphs]` in the reference line)

Exercises

It's time for a demo...

`https://fokg.cs.uni-paderborn.de/`

Exercises

Questions regarding the exercises?

Section 3

Mini-Project

Mini-Project

Overview

- Goal
  - Make use of Semantic Web technologies for a challenging problem
  - Apply approach to real data
- Pass for mini-project iff . . .
  1. ROC AUC above 60% in GERBIL
  2. Documented code on GitHub/GitLab
  3. README with clear instructions for execution
- Note: Project not executable = FAIL

Organization

Mini-Project

Goal

Build a fact-checking engine, which returns a veracity value between 0 (fact is false) and +1 (fact is true) for a given fact with respect to a knowledge graph.

Organization

Mini-Project

Goal

Build a fact-checking engine, which returns a veracity value between 0 (fact is false) and +1 (fact is true) for a given fact with respect to a knowledge graph.

- **Group size**: max. 3 persons
- **Code & documentation**: GitHub/GitLab
- **Suggested steps**
  1. Analyse the given training data
  2. Check how you can make use of the given knowledge base
  3. Fact Checking and Benchmarking
  4. Final submission: July 15th, 2026
- Do **not** wait for the lecture to cover this topic!

Mini-Project

Evaluation overview

Simple three steps to get an evaluation result.

1. TTL file in PANDA containing training data
   (Will be uploaded during the next weeks)
2. Your approach should generate a result file
3. Result files can be uploaded in GERBIL for evaluation

Mini-Project

Training data

The input is represented as RDF statement

```
<http://swc2017.aksw.org/task2/dataset/3892429>
     rdf:type rdf:Statement ;
     rdf:subject <http://dbpedia.org/resource/Venus_Williams> ;
     rdf:predicate <http://dbpedia.org/ontology/birthPlace> ;
     rdf:object
<http://dbpedia.org/resource/Lynwood,_California> .
```

Mini-Project

Result format

Result file format (`result.ttl`)

- One line per fact
  `<Fact-URI> <prop-URI> "value"^^type .`
- Fact-URI is the URI of the `rdf:Statement`
  `http://swc2017.aksw.org/task2/dataset/3892429`
- the prop-URI is always
  `http://swc2017.aksw.org/hasTruthValue`
- the value is the result of your fact checking algorithm (double)
- type of your value should be always
  `<http://www.w3.org/2001/XMLSchema#double>`

Mini-Project

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Possible result for the example:

```
<http://swc2017.aksw.org/task2/dataset/3892429>
  <http://swc2017.aksw.org/hasTruthValue>
  "0.8901"^^<http://www.w3.org/2001/XMLSchema#double> .
```

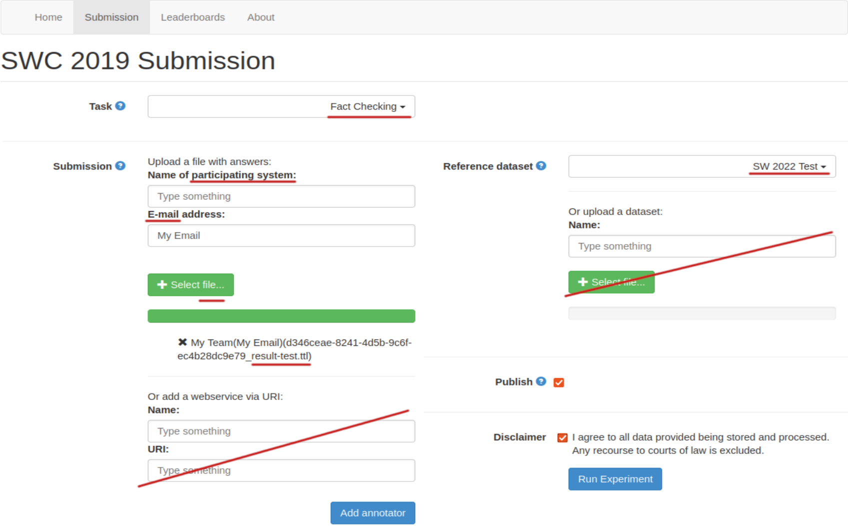
Mini-Project

Evaluate result

1. Go to `http://gerbil-kbc.aksw.org/gerbil/config`.
2. Choose "Fact Checking" as experiment type.
3. Enter your team name and a mail address.
   The address won't be visible. It is used to make sure that you are part of this team.
4. Upload your result file.
5. Choose "FOKG 2024 Train" as reference dataset.
   (Later, "FOKG 2024 Test" for the final evaluation.)
6. Agree to publish your result.
7. Agree to the disclaimer.
8. Submit your result.

Mini-Project

Evaluate result



Mini-Project

Evaluate result

- GERBIL will determine the Area Under Curve (AUC) of your systems ROC curve. This value has the range [0, 1].
- All results will be visible in a leaderboard which can be accessed at `http://gerbil-kbc.aksw.org/gerbil/overview`.
- If you see the message *"The annotator couldn't be loaded"* instead of a result your file has a wrong format (Checking it with an RDF validator might be a good way to go).
- Questions regarding the Mini-Project:
  Umair Qudus: `uqudus@mail.uni-paderborn.de` and
  Yasir Mahmood: `ymahmood@mail.uni-paderborn.de`

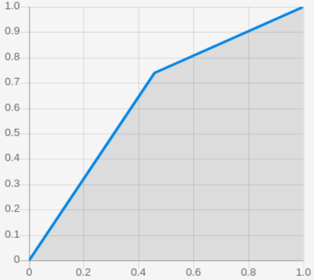
Mini-Project

Evaluate result

Home Submission Leaderboards About

GERBIL Experiment

Experiment URI: http://gerbil-kbc.aksw.org/gerbil/experiment?id=... and http://w3id.org/gerbil/kbc/experiment?id=...
Type: Fact Checking

Annotator	Dataset	Area Under Curve	ROC Curve	avg millis/doc	Timestamp	GERBIL version
...	SW 2022 Train	0,639		0	2022-05-02 15:26:46	0.0.4-SNAPSHOT

Mini-Project

Questions regarding Mini-Project?